17. Найти флуктуации $\overline{\Delta S^2}$, $\overline{\Delta T^2}$, $\overline{\Delta V^2}$, $\overline{\Delta P^2}$, $\overline{\Delta E^2}$, $\overline{\Delta S\Delta P}$, $\overline{\Delta V\Delta P}$, $\overline{\Delta S\Delta T}$, $\overline{\Delta T\Delta V}$, $\overline{\Delta T\Delta P}$, $\overline{\Delta S\Delta V}$.

$w \propto \exp\left(-\frac{\Delta W_{min}}{T}\right)$ – вер-ть возникновения малых флуктуаций

$$w \sim \exp\left(\frac{\Delta S \Delta T - \Delta p \Delta V}{2T}\right)$$

$$dF = -SdT - pdV + \mu dN$$
$$-S = \left(\frac{\partial F}{\partial T}\right)_{V,N} \qquad -p = \left(\frac{\partial F}{\partial V}\right)_{V,N}$$

$$\Delta S = \left(\frac{\partial S}{\partial T}\right)_V \cdot \Delta T + \left(\frac{\partial S}{\partial V}\right)_T \cdot \Delta V = \frac{C_V}{T}\Delta T + \left(\frac{\partial P}{\partial T}\right)_V \cdot \Delta V$$

$$\Delta P = \left(\frac{\partial P}{\partial T}\right)_V \cdot \Delta T + \left(\frac{\partial P}{\partial V}\right)_T \cdot \Delta V$$

$$\Delta S \cdot \Delta T - \Delta p \cdot \Delta V = \left(\frac{C_V}{T}\Delta T + \left(\frac{\partial P}{\partial T}\right)_V \Delta V\right)\Delta T - \Delta V\left(\left(\frac{\partial P}{\partial T}\right)_V \cdot \Delta T + \left(\frac{\partial P}{\partial V}\right)_T \cdot \Delta V\right) = \frac{C_V}{T}(\Delta T)^2 + \left(\frac{\partial P}{\partial T}\right)_V \Delta V \Delta T -$$
$$- \left(\frac{\partial P}{\partial T}\right)_V \Delta V \cdot \Delta T - \left(\frac{\partial P}{\partial V}\right)_T \cdot (\Delta V)^2 = \frac{C_V}{T}(\Delta T)^2 - \left(\frac{\partial P}{\partial V}\right)_T (\Delta V)^2$$

$$w \sim \exp\left(-\frac{C_V}{2T^2}(\Delta T)^2 + \frac{1}{2T}\left(\frac{\partial P}{\partial V}\right)_T \cdot (\Delta V)^2\right)$$

Из стандартного вида распределения Гаусса $\left(w \sim \exp\left(-\frac{x^2}{2\bar{x}} - \frac{y^2}{2\bar{y}}\right)\right)$, получаем:

$$\boxed{\overline{\Delta T^2} = \frac{T^2}{C_V} \qquad \overline{\Delta V^2} = -T\left(\frac{\partial V}{\partial p}\right)_T \qquad \overline{\Delta T \cdot \Delta V} = 0}$$

$$H = E + PV$$
$$dH = TdS + VdP + \mu dN$$
$$T = \left(\frac{\partial H}{\partial S}\right)_{P,N} \qquad V = \left(\frac{\partial H}{\partial P}\right)_{S,N}$$
$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$$

$$\Delta T = \left(\frac{\partial T}{\partial S}\right)_P \cdot \Delta S + \left(\frac{\partial T}{\partial P}\right)_S \cdot \Delta p = \frac{T}{C_p}\Delta S + \left(\frac{\partial V}{\partial S}\right)_P \cdot \Delta p$$

$$\Delta V = \left(\frac{\partial V}{\partial S}\right)_P \cdot \Delta S + \left(\frac{\partial V}{\partial P}\right)_S \cdot \Delta p$$

$$\Delta S \Delta T - \Delta V \cdot \Delta p = \Delta S\left(\frac{T}{C_p}\Delta S + \left(\frac{\partial V}{\partial S}\right)_P \Delta p\right) - \left(\left(\frac{\partial V}{\partial S}\right)_P \cdot \Delta S + \left(\frac{\partial V}{\partial P}\right)_S \Delta p\right) \cdot \Delta p = \frac{T}{C_p}(\Delta S)^2 - \left(\frac{\partial V}{\partial P}\right)_S \cdot \Delta p^2$$

$$w \sim \exp\left(-\frac{1}{2C_p}(\Delta S)^2 + \frac{1}{2T}\left(\frac{\partial V}{\partial P}\right)_S (\Delta p)^2\right)$$

$$\Rightarrow \boxed{\overline{\Delta S^2} = C_p \qquad \overline{\Delta p^2} = -T\left(\frac{\partial P}{\partial V}\right)_S \qquad \overline{\Delta S \cdot \Delta p} = 0}$$

$$\overline{\Delta S \cdot \Delta T} = \overline{\left(\frac{C_V}{T}\Delta T + \left(\frac{\partial P}{\partial T}\right)_V \cdot \Delta V\right)\Delta T} = \frac{C_V}{T}\overline{\Delta T^2} + \left(\frac{\partial P}{\partial T}\right)_V \cdot \overline{\Delta V \cdot \Delta T} = \frac{C_V}{T} \cdot \frac{T^2}{C_V} = T$$

$$\overline{\Delta V \cdot \Delta p} = \overline{\Delta T\left(\left(\frac{\partial P}{\partial T}\right)_V \cdot \Delta T + \left(\frac{\partial P}{\partial V}\right) \cdot \Delta V\right)} = \left(\frac{\partial P}{\partial T}\right)_V \cdot \overline{\Delta T^2} + \left(\frac{\partial P}{\partial V}\right)_T \cdot \overline{\Delta T \cdot \Delta V} = \frac{T^2}{C_V} \cdot \left(\frac{\partial P}{\partial T}\right)_V$$

$$\overline{\Delta S \cdot \Delta V} = \overline{\left(\frac{C_V}{T} \cdot \Delta T + \left(\frac{\partial P}{\partial T}\right)_V \cdot \Delta V\right) \cdot \Delta V} = \frac{C_V}{T} \cdot \overline{\Delta T \cdot \Delta V} + \left(\frac{\partial P}{\partial T}\right)_V \cdot \overline{\Delta V^2} = -T\left(\frac{\partial P}{\partial T}\right)_V \cdot \left(\frac{\partial V}{\partial P}\right)_T = T\left(\frac{\partial V}{\partial T}\right)_P$$

$$\Delta E = \left(\frac{\partial E}{\partial T}\right)_V \cdot \Delta T + \left(\frac{\partial E}{\partial V}\right)_T \cdot \Delta V = C_V \Delta T + \left(T\left(\frac{\partial P}{\partial T}\right)_V - p\right)\Delta V$$

$$\overline{\Delta E^2} = C_V^2\,\overline{\Delta T^2} + \left(T\left(\frac{\partial P}{\partial T}\right)_V - p\right)^2 \cdot \overline{\Delta V^2} = C_V \frac{T^2}{C_V} + \left(T\left(\frac{\partial P}{\partial T}\right)_V - p\right)^2 \cdot \left(-T \cdot \left(\frac{\partial V}{\partial P}\right)_T\right) = T^2 C_V - T\left(T\left(\frac{\partial P}{\partial T}\right)_V -\right.$$
$$\left. - p\right)^2 \cdot \left(\frac{\partial V}{\partial P}\right)_T$$